

Your grade: 80%

Your latest: 80% • Your highest: 80% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed a good understanding of machine learning concepts and activation functions, but there is room for growth in understanding the advantages of PyTorch. Before retrying this assignment, we recommend focusing on this key area:

- PyTorch Introduction:** In Question 1, your understanding of the primary advantage of using PyTorch for deep learning projects was incorrect. Try reviewing [this item](#) to refresh your memory.

Good job on the rest of the questions!

Resume assignment



1. What is the primary advantage of using PyTorch for deep learning projects?

0 / 1 point

- ☐ Dynamic computational graphs
- ☐ Better GPU support than TensorFlow
- ☒ Larger community and support
- ☐ Simpler syntax for linear algebra

Incorrect

Incorrect. While PyTorch has a strong community, this isn't its primary advantage over other frameworks.

2. What differentiates machine learning from classical programming?

1 / 1 point

- ☐ Machine learning models do not need any data, while classical programming requires large datasets
- ☐ Machine learning models are always more accurate than classical programming
- ☐ Machine learning is only used in academic research, while classical programming is used in commercial applications
- ☒ Machine learning models learn from data, while classical programming relies on explicit instructions

Correct

Correct! Machine learning models use data to improve their performance, whereas classical programming relies on predefined rules.

3. Which of the following are correct steps in developing a machine learning model? Select all that apply.

1 / 1 point

☒ Data collection

Correct

Correct! Collecting relevant data is the first step in developing any machine learning model.

☒ Model training

Correct

Correct! Training the model on collected data is a crucial step in the development process.

☒ Feature engineering

Correct

Correct! Feature engineering involves selecting and transforming variables to improve the model's performance.

☐ Writing complex rules

☒ Model deployment

Correct

Correct! Deploying the model is the final step to make it available for use in real-world applications.

4. Select the activation functions that are non-linear and commonly used in neural networks.

1 / 1 point

☐ Step Function

☒ ReLu (Rectified Linear Unit)

Correct

Correct! ReLu is non-linear and commonly used in hidden layers.

☒ Hyperbolic Tangent (Tanh)

Correct

Correct! Tanh is a non-linear activation function and is used in neural networks.

☒ Sigmoid

Correct

Correct! The Sigmoid function is non-linear and was historically used in neural networks.

☒ Leaky ReLu

Correct

Correct! Leaky ReLu is a variation of ReLu that aims to solve the dying ReLu problem.

☐ Linear Function

5. Which loss function is most suitable for binary classification problems?

1 / 1 point

- ☐ Multi-label Cross Entropy
- ☐ Hinge Loss
- ☐ Mean Squared Error
- ☒ Binary Cross Entropy

Correct

Correct! Binary Cross Entropy is designed for binary classification tasks.